



At a glance: India's regulatory and institutional framework

A mix of legacy atomic regulations and modern industrial policy shapes India's rare earth strategy:

- **Atomic Energy Act (1962).** Monazite and thorium-bearing sands remain under tight state control, limiting private exploration and processing
- **MMDR Amendment (2023).** Centralises auctions for 24 critical minerals; allows exploration licences for 29 more; streamlines private and foreign participation
- **National Critical Minerals Mission (NCMM, 2025).** Targets 1200 GSI exploration projects, ₹15 billion recycling incentives, and 1000 patents by FY 2030–31; promotes PSU-industry collaboration
- **National Mineral Exploration Trust (NMET).** Provides 100% funding for notified exploration agencies; 50% reimbursement for private explorers; 25% incentives for deep-seated 'G4' minerals
- **Critical Mineral Auctions.** 34 blocks auctioned across five tranches, including offshore resources; GSI leads mapping and evaluation
- **Tariff & Customs Relief.** Duties eliminated for critical minerals such as cobalt powder and battery scrap to aid downstream manufacturing
- **Magnet Manufacturing Incentive Scheme (₹73 billion).** Supports 6000 tonnes of domestic magnet capacity by 2030; part of 'China-plus-one' diversification
- **Critical Mineral Stockpile Programme (planned).** To establish two-month reserves of essential rare earth elements to safeguard industry supply
- **Institutional Pillars.** Ministry of Mines, Department of Atomic Energy (DAE), IREL (India) Ltd, GSI, AMD, and MEDEPC coordinate exploration, R&D, and policy implementation

They aim to secure continuity, not transformation. The larger challenge and opportunity lie in developing technologies and materials that can replace rare earths in electric vehicles, wind turbines, and electronics.

Globally, research into alternatives is accelerating. Toyota and Hitachi have made progress on heavy rare-earth-free magnet designs, while the US and EU laboratories are exploring iron-nitride and aluminium-nickel-cobalt alloys. According to the US Department of Energy, "substitution and recycling together could reduce global dependence on rare earths by up to 25% by 2040." Yet, India's industrial and R&D ecosystem remains under-prepared for such innovation-led diversification.

Domestic R&D expenditure accounts for only 0.7% of GDP, one of the lowest rates among major economies (UNESCO, 2024). Most Indian electric motor manufacturers still rely on imported neodymium-iron-boron magnets, with no large-scale substitute technology under

commercialisation. Even within the Production Linked Incentive (PLI) schemes for electronics and EVs, no dedicated track exists for material substitution research or magnet-free motor innovation—areas where global players like Advanced Electric Machines (UK) and Niron Magnetics (US) are already scaling up.

A 2024 report by the Council on Energy, Environment, and Water (CEEW) argues that India's current approach resembles a resource-security model, not a technology-sovereignty strategy. Securing ores and refining capacity may reduce vulnerability to Chinese supply shocks, but without investment in fundamental materials science, the dependency simply moves upstream.

India's next step, therefore, must go beyond mining licences and magnet plants. It requires strategic R&D funding, university-industry collaboration, and incentives for research on rare-earth-free motors and semiconductors. True resilience will come not just from controlling the supply, but from making the supply optional.

A pathway of freedom from the rare earth magnet

That shift is already beginning to take shape on the ground. Across India's tech hubs, from Bengaluru and Chennai to Pune, homegrown innovators and startups are quietly reimagining motor design. Moving away from the policy complications of rare earth procurement, they are asking a simple question: If we cannot reliably buy those magnets, why not build motors that don't need them?

Using ferrite magnets and combinations of aluminium, copper, and steel, several ventures are achieving magnetic performance comparable to that of neodymium-based systems. These early experiments signal a turn, one in which India's path to rare earth resilience may run not only through mines and factories, but also through labs and startups that make rare earth elements less essential to begin with.

Instead of importing neodymium magnets, the new generation of rare earth-free motors comes with the same magnetic power. "We are not chasing rare materials anymore," says Bhaktha Keshavachar, Founder and CEO of Bengaluru-based startup Chara Technologies.

Chara's architecture replaces rare earth magnets with rotors and software-controlled magnetic fields. Their synchronous reluctance motors are designed for two-wheelers, three-wheelers, and small commercial vehicles, with the most substantial traction in three-wheelers. Their applications are expanding to farm loaders, forklifts, and campus vehicles, supported by a modular 48-400V platform where only the shaft and mounting details differ.

The company has raised ₹520 million in funding and is scaling production from 20,000 to 100,000 electric three-wheelers, loaders, and other light commercial vehicles per year. "Once you crack the design,